

NYC TLC Data Lakehouse

Schema Reference

AWS Glue Data Catalog · Account 861276091613
Generated: 2026-05-22

Overview

Database	Layer	Tables	Storage
nyc_tlc_gold_marts	Gold marts	4	Parquet (Athena results)
nyc_tlc_gold	Gold	3	Delta Lake + Parquet
nyc_tlc_silver	Silver	2	Delta Lake
nyc_tlc_gold_staging	Staging	2	View (dbt)
nyc_tlc_gold_intermediate	Intermediate	1	View (dbt)

Partition key warning: hourly_kpis in nyc_tlc_gold has pickup_year and pickup_month as VARCHAR. Always use string literals when filtering — pickup_year = '2024', pickup_month = '01' — or cast: CAST(pickup_year AS INTEGER) = 2024.

Gold Marts layer

`nyc_tlc_gold_marts`

Star-schema mart layer. `fact_trips` joins to `dim_location` on `pickup/dropoff_location_id` and to `dim_datetime` on `pickup_date`. `mart_hourly_kpi` is a pre-aggregated summary with denormalised dimension attributes. Use this layer for all analytical queries.

fact_trips

Format: Parquet **Location:** `s3://nyc-tlc-athena-results-dev-861276091613/results/tables/nyc_tlc_gold_marts/fact_trips/...`

Core trip fact table with enriched payment type and rate code labels, speed, and all financial columns.

#	Column name	Data type	Notes
1	<code>trip_id</code>	<code>varchar(36)</code>	PK / Partition key · Surrogate UUID
2	<code>pickup_location_id</code>	<code>int</code>	→ <code>dim_location.location_id</code>
3	<code>dropoff_location_id</code>	<code>int</code>	→ <code>dim_location.location_id</code>
4	<code>pickup_at</code>	<code>timestamp</code>	Pickup timestamp
5	<code>dropoff_at</code>	<code>timestamp</code>	
6	<code>pickup_date</code>	<code>date</code>	→ <code>dim_datetime.date_day</code>
7	<code>pickup_hour</code>	<code>bigint</code>	
8	<code>cab_type</code>	<code>varchar(6)</code>	'yellow' or 'green'
9	<code>vendor_id</code>	<code>int</code>	
10	<code>passenger_count</code>	<code>int</code>	
11	<code>store_and_fwd_flag</code>	<code>string</code>	
12	<code>payment_type</code>	<code>varchar(11)</code>	'Credit card', 'Cash', 'No charge', etc.
13	<code>rate_code</code>	<code>varchar(21)</code>	'Standard rate', 'JFK', 'Newark', etc.
14	<code>trip_distance_miles</code>	<code>double</code>	
15	<code>trip_duration_seconds</code>	<code>bigint</code>	
16	<code>avg_speed_mph</code>	<code>double</code>	
17	<code>fare_amount</code>	<code>double</code>	
18	<code>extra</code>	<code>double</code>	
19	<code>mta_tax</code>	<code>double</code>	
20	<code>tip_amount</code>	<code>double</code>	
21	<code>tolls_amount</code>	<code>double</code>	
22	<code>improvement_surcharge</code>	<code>double</code>	
23	<code>congestion_surcharge</code>	<code>double</code>	
24	<code>airport_fee</code>	<code>double</code>	

#	Column name	Data type	Notes
25	cbd_congestion_fee	double	
26	total_amount	double	
27	is_airport_trip	boolean	

dim_location

Format: Parquet **Location:** s3://nyc-tlc-athena-results-dev-861276091613/results/tables/nyc_tlc_gold_marts/dim_location/...

Taxi zone dimension enriched with borough grouping and airport/Manhattan CBD flags.

#	Column name	Data type	Notes
1	location_id	int	PK / Partition key
2	zone_name	string	
3	borough	string	
4	service_zone	string	
5	borough_group	varchar(14)	Grouped borough label
6	is_airport	boolean	
7	is_manhattan_cbd	boolean	

dim_datetime

Format: Parquet **Location:** s3://nyc-tlc-athena-results-dev-861276091613/results/tables/nyc_tlc_gold_marts/dim_datetime/...

Date dimension spanning the full trip date range with US holiday support.

#	Column name	Data type	Notes
1	date_day	date	PK / Partition key
2	year	bigint	
3	month	bigint	
4	day_of_month	bigint	
5	day_of_week	bigint	1 = Monday ... 7 = Sunday
6	week_of_year	bigint	
7	quarter	bigint	
8	year_month	string	e.g. '2024-03'
9	day_name	string	
10	month_name	string	
11	year_quarter	string	e.g. '2024-Q1'

#	Column name	Data type	Notes
12	is_weekend	boolean	
13	us_holiday_name	varchar(16)	
14	is_us_holiday	boolean	

mart_hourly_kpi

Format: Parquet **Location:** s3://nyc-tlc-athena-results-dev-861276091613/results/tables/nyc_tlc_gold_marts/mart_hourly_kpi/...

Pre-aggregated hourly KPI mart with denormalised dimension attributes. Designed for fast BI queries without joins.

#	Column name	Data type	Notes
1	pickup_date	date	
2	pickup_hour	bigint	
3	cab_type	varchar(6)	
4	borough_group	varchar(14)	
5	is_airport	boolean	
6	is_manhattan_cbd	boolean	
7	is_weekend	boolean	
8	is_us_holiday	boolean	
9	day_name	string	
10	month_name	string	
11	year_month	string	
12	year_quarter	string	
13	trip_count	bigint	
14	total_passengers	bigint	
15	total_distance_miles	double	
16	avg_distance_miles	double	
17	avg_duration_minutes	double	
18	median_duration_minutes	double	
19	avg_speed_mph	double	
20	total_fare	double	
21	total_tips	double	
22	total_revenue	double	
23	avg_fare	double	

#	Column name	Data type	Notes
24	avg_tip_rate	double	
25	airport_trips	bigint	
26	revenue_per_mile	double	
27	rolling_7d_trips	double	
28	rolling_7d_revenue	double	
29	unique_pickup_zones	bigint	

Gold layer

nyc_tlc_gold

Delta Lake gold tables exposed to Athena via symlink manifests. fact_trips and hourly_kpis are Delta Lake tables; taxi_zone_lookup is a static reference stored in the Athena results bucket.

fact_trips

Format: Delta Lake / Parquet **Location:** s3://nyc-tlc-datalake-dev-861276091613/gold/fact_trips/_symlink_format_manifest

Gold Delta Lake trip fact. Column names use raw TLC conventions (pulocationid, dolocationid).

#	Column name	Data type	Notes
1	vendorid	bigint	
2	pickup_datetime	timestamp	
3	dropoff_datetime	timestamp	
4	passenger_count	bigint	
5	trip_distance	double	
6	ratecodeid	bigint	
7	store_and_fwd_flag	string	
8	pulocationid	bigint	
9	dolocationid	bigint	
10	payment_type	bigint	
11	fare_amount	double	
12	extra	double	
13	mta_tax	double	
14	tip_amount	double	
15	tolls_amount	double	
16	improvement_surcharge	double	
17	total_amount	double	
18	congestion_surcharge	double	
19	airport_fee	double	
20	cbd_congestion_fee	double	

hourly_kpis

Format: Delta Lake / Parquet **Location:** s3://nyc-tlc-datalake-dev-861276091613/gold/hourly_kpis/_symlink_format_manifest

Gold hourly KPI table. IMPORTANT: pickup_year and pickup_month are VARCHAR partition keys. Always use string literals when filtering: pickup_year = '2024', pickup_month = '01'.

#	Column name	Data type	Notes
1	pickup_date	date	
2	pickup_hour	int	
3	pickup_borough	string	
4	trip_count	bigint	
5	total_revenue	double	
6	avg_fare	double	
7	avg_distance_mi	double	
8	avg_duration_min	double	
9	avg_tip_rate	double	
10	airport_trips	bigint	
11	weekend_trips	bigint	
12	unique_pickup_zones	bigint	
13	pickup_datetime_hour	timestamp	
14	rolling_7d_trips	double	
15	rolling_7d_revenue	double	
16	revenue_per_mile	double	
17	pickup_year	string	PK / Partition key · Partition (VARCHAR) — filter: pickup_year = '2024'
18	pickup_month	string	PK / Partition key · Partition (VARCHAR) — filter: pickup_month = '01'

taxi_zone_lookup

Format: Parquet **Location:** s3://nyc-tlc-athena-results-dev-861276091613/results/tables/nyc_tlc_gold/taxi_zone_lookup/...

Static taxi zone reference table. 265 rows covering all NYC TLC zones.

#	Column name	Data type	Notes
1	locationid	int	PK / Partition key
2	borough	string	
3	zone	string	
4	service_zone	string	

Silver layer

nyc_tlc_silver

Cleaned, typed Delta Lake tables for yellow and green taxi trips. These are the source-of-truth tables for all downstream dbt models.

yellow

Format: Delta Lake / Parquet **Location:** s3://nyc-tlc-datalake-dev-861276091613/silver/yellow/_symlink_format_manifest

Cleaned yellow taxi trip records. Preserves raw TLC column names with corrected types.

#	Column name	Data type	Notes
1	vendorid	bigint	
2	pickup_datetime	timestamp	
3	dropoff_datetime	timestamp	
4	passenger_count	bigint	
5	trip_distance	double	
6	ratecodeid	bigint	
7	store_and_fwd_flag	string	
8	pulocationid	bigint	
9	dolocationid	bigint	
10	payment_type	bigint	
11	fare_amount	double	
12	extra	double	
13	mta_tax	double	
14	tip_amount	double	
15	tolls_amount	double	
16	improvement_surcharge	double	
17	total_amount	double	
18	congestion_surcharge	double	
19	airport_fee	double	
20	cbd_congestion_fee	double	

green

Format: Delta Lake / Parquet **Location:** s3://nyc-tlc-datalake-dev-861276091613/silver/green/_symlink_format_manifest

Cleaned green taxi trip records. Identical schema to silver.yellow.

#	Column name	Data type	Notes
1	vendorid	bigint	
2	pickup_datetime	timestamp	
3	dropoff_datetime	timestamp	
4	passenger_count	bigint	
5	trip_distance	double	
6	ratecodeid	bigint	
7	store_and_fwd_flag	string	
8	pulocationid	bigint	
9	dolocationid	bigint	
10	payment_type	bigint	
11	fare_amount	double	
12	extra	double	
13	mta_tax	double	
14	tip_amount	double	
15	tolls_amount	double	
16	improvement_surcharge	double	
17	total_amount	double	
18	congestion_surcharge	double	
19	airport_fee	double	
20	cbd_congestion_fee	double	

Staging layer

nyc_tlc_gold_staging

dbt staging views over the silver layer. Rename and cast columns to project naming conventions (snake_case, _at suffix for timestamps, _miles suffix for distance). Input for intermediate models.

stg_yellow_trips

Format: View

dbt staging model over silver.yellow. Renames pulocationid → pickup_location_id, adds pickup_date and pickup_hour derived columns.

#	Column name	Data type	Notes
1	vendor_id	int	
2	rate_code_id	int	
3	pickup_location_id	int	
4	dropoff_location_id	int	
5	payment_type_id	int	
6	store_and_fwd_flag	string	
7	passenger_count	int	
8	pickup_at	timestamp	
9	dropoff_at	timestamp	
10	pickup_date	date	
11	pickup_hour	bigint	
12	trip_distance_miles	double	
13	trip_duration_seconds	bigint	
14	fare_amount	double	
15	extra	double	
16	mta_tax	double	
17	tip_amount	double	
18	tolls_amount	double	
19	improvement_surcharge	double	
20	total_amount	double	
21	congestion_surcharge	double	
22	airport_fee	double	
23	cbd_congestion_fee	double	

stg_green_trips**Format:** View*dbt staging model over silver.green. Identical schema to stg_yellow_trips.*

#	Column name	Data type	Notes
1	vendor_id	int	
2	rate_code_id	int	
3	pickup_location_id	int	
4	dropoff_location_id	int	
5	payment_type_id	int	
6	store_and_fwd_flag	string	
7	passenger_count	int	
8	pickup_at	timestamp	
9	dropoff_at	timestamp	
10	pickup_date	date	
11	pickup_hour	bigint	
12	trip_distance_miles	double	
13	trip_duration_seconds	bigint	
14	fare_amount	double	
15	extra	double	
16	mta_tax	double	
17	tip_amount	double	
18	tolls_amount	double	
19	improvement_surcharge	double	
20	total_amount	double	
21	congestion_surcharge	double	
22	airport_fee	double	
23	cbd_congestion_fee	double	

Intermediate layer

`nyc_tlc_gold_intermediate`

UNION ALL of `stg_yellow_trips` and `stg_green_trips` with a surrogate `trip_id`. Used as the single input to all gold and mart models.

int_trips_unioned

Format: View

Union of all taxi staging tables. Adds surrogate `trip_id` key. This is the single source for `fact_trips` in the mart layer.

#	Column name	Data type	Notes
1	<code>trip_id</code>	string	PK / Partition key · Surrogate UUID — SHA256 hash of source fields
2	<code>vendor_id</code>	int	
3	<code>rate_code_id</code>	int	
4	<code>pickup_location_id</code>	int	
5	<code>dropoff_location_id</code>	int	
6	<code>payment_type_id</code>	int	
7	<code>store_and_fwd_flag</code>	string	
8	<code>passenger_count</code>	int	
9	<code>pickup_at</code>	timestamp	
10	<code>dropoff_at</code>	timestamp	
11	<code>pickup_date</code>	date	
12	<code>pickup_hour</code>	bigint	
13	<code>trip_distance_miles</code>	double	
14	<code>trip_duration_seconds</code>	bigint	
15	<code>fare_amount</code>	double	
16	<code>extra</code>	double	
17	<code>mta_tax</code>	double	
18	<code>tip_amount</code>	double	
19	<code>tolls_amount</code>	double	
20	<code>improvement_surcharge</code>	double	
21	<code>total_amount</code>	double	
22	<code>congestion_surcharge</code>	double	
23	<code>airport_fee</code>	double	

#	Column name	Data type	Notes
24	cbd_congestion_fee	double	